

CHAPTER 7

IP Addressing and Subnetting

IPv4 · classes · private and special addresses · subnet masks and CIDR · subnetting and VLSM · NAT · IPv4 header and fragmentation · IPv6 · routing basics

QUICK TRICK

Learn the fast tricks first (details below):

1. **Class by first octet:** A = 1-126 · B = 128-191 · C = 192-223 · D = 224-239 (multicast) · E = 240-255. 127 = loopback.
2. **Hosts in a subnet** = $2^h - 2$ (h = host bits). **Block size** = 256 – mask octet.
3. **Network address** = all host bits 0. **Broadcast address** = all host bits 1. **Usable hosts** lie between them.
4. **Private ranges:** 10.0.0.0/8 · 172.16.0.0/12 (172.16-172.31) · 192.168.0.0/16. **IPv6** = 128 bits, written in hex.

7.1 What is an IP address?

● LEVEL 1 · BEGINNER

The problem. A MAC address (Chapter 3) is fixed in the card and says nothing about **where** the device is. A router cannot keep a list of billions of MAC addresses. **The idea.** An **IP address** is a **logical address** given to every network **interface**. It has two parts: a **network part** (which network) and a **host part** (which device in that network), like a **PIN code + house number**. Routers look only at the network part.

IPv4 address = **32 bits**, written as **4 octets** (bytes) in **dotted decimal**, each octet 0 to 255. Example: **192.168.1.10**. Total addresses = 2^{32} = **4 294 967 296** (about 4.3 billion), which is not enough today, so IPv6 exists.

Converting between binary and decimal

Each octet has 8 bits with weights **128, 64, 32, 16, 8, 4, 2, 1**.

Weight	128	64	32	16	8	4	2	1
172 =	1	0	1	0	1	1	0	0
10 =	0	0	0	0	1	0	1	0

Example. 172 = 128 + 32 + 8 + 4 → 10101100. 10 = 8 + 2 → 00001010. So **192.168.1.10** = 11000000.10101000.00000001.00001010 (192 = 11000000, 168 = 10101000, 1 = 00000001, 10 = 00001010).

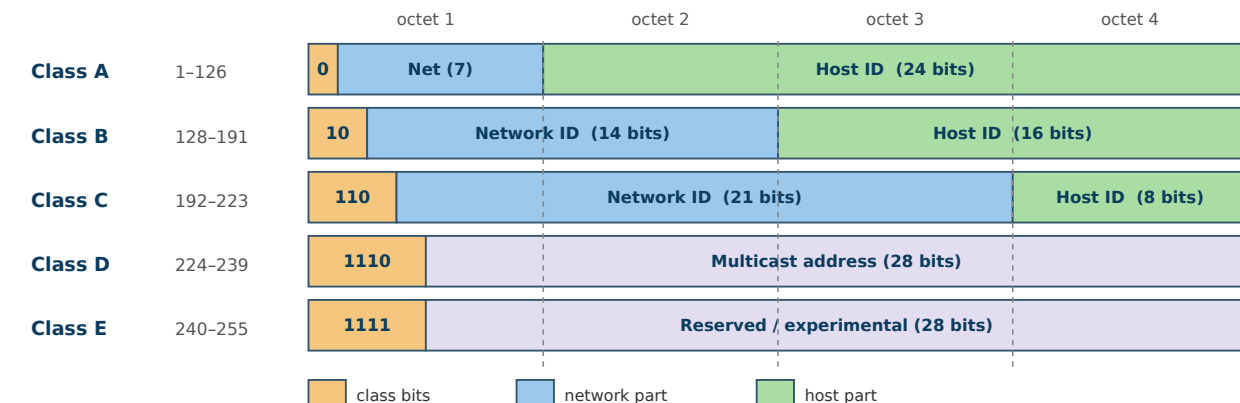
QUICK TRICK

To make decimal → binary quickly, subtract weights from the left: if the weight fits, write 1 and subtract it; if not, write 0. Useful numbers: **128 = 10000000**, **192 = 11000000**, **224 = 11100000**, **240 = 11110000**, **248 = 11111000**, **252 = 11111100**, **254 = 11111110**, **255 = 11111111**.

7.2 Classful addressing: classes A to E

● LEVEL 1 · BEGINNER

Originally the 32 bits were divided into **five classes**. The **first few bits** of the address show the class; the class decides where the network part ends.



Class	First bits	First octet	Format	Default mask	Networks	Hosts per network
A	0	1 - 126	N.H.H.H	255.0.0.0 (/8)	$2^7 - 2 = 126$	$2^{24} - 2 = 16\,777\,214$
B	10	128 - 191	N.N.H.H	255.255.0.0 (/16)	$2^{14} = 16\,384$	$2^{16} - 2 = 65\,534$
C	110	192 - 223	N.N.N.H	255.255.255.0 (/24)	$2^{21} = 2\,097\,152$	$2^8 - 2 = 254$
D	1110	224 - 239	—	—	—	Multicast groups
E	1111	240 - 255	—	—	—	Reserved for experiments

BPSC POINT

Class A has **126** networks (not 128) because network **0** (this network) and network **127** (loopback) are reserved. Hosts are $2^h - 2$ because the all-0 host part is the **network address** and the all-1 host part is the **broadcast address**.

Special and private addresses

Address	Meaning
Network address	Host part all 0s, for example 192.168.1.0; names the network itself
Directed broadcast	Host part all 1s, for example 192.168.1.255; reaches all hosts of that network
Limited broadcast	255.255.255.255; all hosts on the local network, never forwarded by routers
This host	0.0.0.0; used by a host that does not yet know its address (for example before DHCP)
Loopback	127.0.0.0/8, mainly 127.0.0.1 (localhost); traffic stays inside the computer
APIPA / link-local	169.254.0.0/16; self-assigned when no DHCP server answers
Multicast	224.0.0.0/4 (class D)
Default route	0.0.0.0/0; matches every destination

Private range (RFC 1918)	From - To	Prefix	Class equivalent	Addresses
10.0.0.0	10.0.0.0 - 10.255.255.255	/8	1 Class A	16 777 216
172.16.0.0	172.16.0.0 - 172.31 .255.255	/12	16 Class B	1 048 576
192.168.0.0	192.168.0.0 - 192.168.255.255	/16	256 Class C	65 536

Private addresses are used inside homes and organisations and are **not routed on the Internet**. Everything else is **public** (must be unique worldwide, given by ISPs and regional registries such as APNIC for Asia-Pacific).

COMMON MISTAKE

The private range is 172.**16**.0.0 to 172.**31**.255.255, **not** all of 172.x.x.x. For example 172.32.1.1 and 172.15.1.1 are **public**.

Finding the network ID and host ID in a classful address

Example. 130.45.12.9. First octet 130 → **Class B** (N.N.H.H, mask 255.255.0.0).

Network address = **130.45.0.0**. Host part = 0.0.12.9. Broadcast = **130.45.255.255**. Usable hosts = 130.45.0.1 to 130.45.255.254 (65 534 hosts).

7.3 Subnet mask, CIDR and prefix length

● LEVEL 2 · EXAM

A **subnet mask** is a 32-bit number with **1s for the network part** and **0s for the host part**. A router does a bitwise **AND** of the address and the mask to get the network address.

	Octet 1	Octet 2	Octet 3	Octet 4
IP address 192.168.10.77	11000000	10101000	00001010	01001101
Mask 255.255.255.192	11111111	11111111	11111111	11000000
AND = network	11000000 (192)	10101000 (168)	00001010 (10)	01000000 (64)

Network = **192.168.10.64**. The host part is the last 6 bits; broadcast = all 6 host bits 1 = $64 + 63 = 192.168.10.127$. Usable hosts: **192.168.10.65 to 192.168.10.126** ($2^6 - 2 = 62$ hosts).

CIDR (Classless Inter-Domain Routing, 1993) drops the fixed classes. The network part can be **any length**, written as a **prefix length** after a slash: **192.168.10.64/26** means the first 26 bits are the network part. It also allows **route aggregation** (supernetting), which keeps routing tables small.

Prefix	Subnet mask	Host bits	Block size	Usable hosts
/8	255.0.0.0	24	—	16 777 214
/16	255.255.0.0	16	—	65 534
/19	255.255.224.0	13	32 (3rd octet)	8 190
/20	255.255.240.0	12	16 (3rd octet)	4 094
/21	255.255.248.0	11	8 (3rd octet)	2 046
/22	255.255.252.0	10	4 (3rd octet)	1 022
/23	255.255.254.0	9	2 (3rd octet)	510
/24	255.255.255.0	8	256	254
/25	255.255.255.128	7	128	126
/26	255.255.255.192	6	64	62
/27	255.255.255.224	5	32	30
/28	255.255.255.240	4	16	14
/29	255.255.255.248	3	8	6
/30	255.255.255.252	2	4	2
/31	255.255.255.254	1	2	2 (point-to-point links, no network or broadcast address)
/32	255.255.255.255	0	1	1 (a single host)

QUICK TRICK

Block-size method (fast, no binary): **block size = 256 – mask octet**. Example /26: mask octet 192 → block = 64. Subnets start at 0, 64, 128, 192.

To find the network of a host, take the host octet and round **down** to a multiple of the block size: 77 → 64. Broadcast = next network – 1 = 127.

Mask octet values: 128, 192, 224, 240, 248, 252, 254, 255 (these are 1, 2, 3, 4, 5, 6, 7, 8 ones).

Supernetting example. Four networks 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24 are contiguous, and 0 is a multiple of 4. They aggregate into **192.168.0.0/22** (block size 4 in the third octet), so one routing entry replaces four.

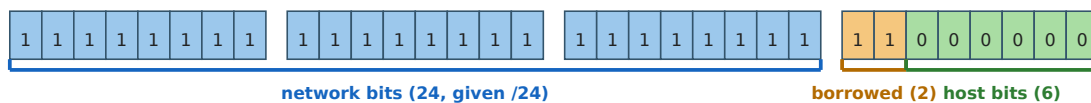
7.4 Subnetting

● LEVEL 2 · EXAM

Subnetting divides one network into several smaller **subnets** by **borrowing bits from the host part**. It reduces broadcast traffic, improves security and uses addresses more wisely.

Step	What to do
1	Find the number of subnets needed, n. Borrow b bits so that $2^b \geq n$.
2	New prefix = old prefix + b. Write the new mask.
3	Host bits left h = 32 – new prefix. Usable hosts = $2^h - 2$.
4	Block size = 256 – mask octet (in the octet where the split happens). List the subnets by adding the block size.
5	For each subnet: network = start, broadcast = next start – 1, hosts = from network + 1 to broadcast – 1.

192.168.1.0/24 split into 4 subnets: borrow 2 host bits → new prefix /26



mask 255.255.255.192 = 26 ones followed by 6 zeros

Example 1. Split 192.168.1.0/24 into 4 equal subnets. $2^2 = 4$, so borrow 2 bits: new prefix **/26**, mask **255.255.255.192**, block size 64, host bits 6, hosts = $2^6 - 2 = 62$.

Subnet	Network address	First host	Last host	Broadcast
1	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193	192.168.1.254	192.168.1.255

Example 2 (Class B). Network 172.16.0.0/16 must have at least **50 subnets**, each with at least **1000 hosts**. $2^6 = 64 \geq 50$, so borrow **6 bits**: new prefix **/22**, mask **255.255.252.0**, host bits 10, hosts = $2^{10} - 2 = 1022 \geq 1000$ ✓. Block size = 256 – 252 = 4 in the third octet. Subnets: 172.16.0.0, 172.16.4.0, 172.16.8.0, ... up to 172.16.252.0 (64 subnets).

Example 3 (which subnet?). Host 10.1.37.200/21. Mask 255.255.248.0, block size 8 in the third octet. $37 \div 8 = 4.6$, so $4 \times 8 = 32$. Network = **10.1.32.0**, broadcast = **10.1.39.255**, usable range 10.1.32.1 to 10.1.39.254, hosts = $2^{11} - 2 = 2046$.

COMMON MISTAKE

- Forgetting the **– 2** for hosts (network and broadcast addresses cannot be given to hosts).
- Counting subnets as $2^b - 2$. In old classful rules the first (subnet zero) and last subnets were avoided; with **CIDR / modern routers all 2^b subnets are usable**. Follow the rule given in the question.
- Giving the broadcast address of one subnet to a host, or the network address to a host: **192.168.1.64** and **192.168.1.127** are not valid hosts in the /26 above.

VLSM: Variable Length Subnet Mask

In plain subnetting all subnets have the same size, which wastes addresses (a link with 2 routers gets 62 spare addresses). **VLSM** gives **each subnet its own size**. Rule: **sort the requirements from largest to smallest** and allocate in that order, starting at the beginning of the block.

Example. Block 192.168.10.0/24. Needs: LAN A = 100 hosts, LAN B = 50 hosts, LAN C = 20 hosts, one WAN link = 2 hosts.

100 → 7 host bits (126) → /25. 50 → 6 bits (62) → /26. 20 → 5 bits (30) → /27. 2 → 2 bits (2) → /30.

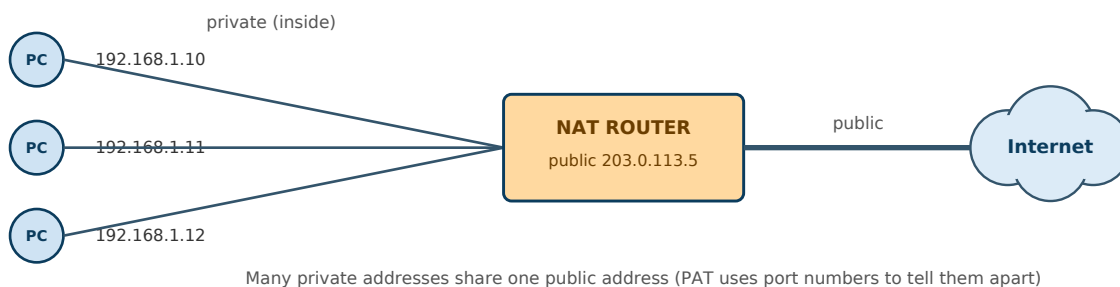
Subnet	Need	Prefix	Network	Usable hosts	Broadcast
LAN A	100	/25	192.168.10.0	.1 - .126	192.168.10.127
LAN B	50	/26	192.168.10.128	.129 - .190	192.168.10.191
LAN C	20	/27	192.168.10.192	.193 - .222	192.168.10.223
WAN link	2	/30	192.168.10.224	.225 - .226	192.168.10.227

Used = 128 + 64 + 32 + 4 = 228 addresses; **192.168.10.228 - .255** remain free for the future.

7.5 NAT: Network Address Translation

● LEVEL 2 · EXAM

Private addresses cannot go on the Internet, and public IPv4 addresses are scarce. A **NAT router** rewrites the source address of outgoing packets from a **private** address to its own **public** address, and reverses it for the replies. This lets a whole home or office share one public IP.



Type	How it works
Static NAT	One private address ↔ one fixed public address (for a server)
Dynamic NAT	Private addresses use a pool of public addresses, one at a time
PAT (NAT overload)	Many private addresses share one public address; the router uses port numbers to tell the connections apart

Direction	Source (before → after)	Destination
Outgoing	192.168.1.10 : 51000 → 203.0.113.5 : 62000	93.184.216.34 : 80
Reply	93.184.216.34 : 80	203.0.113.5 : 62000 → 192.168.1.10 : 51000

Advantages: saves public addresses, hides the inside network. **Disadvantages:** breaks end-to-end addressing, adds delay, makes some protocols (peer-to-peer, VoIP) harder. IPv6 removes the need for NAT.

7.6 IPv4 header and fragmentation

● LEVEL 3 · BPSC / PGT

Field	Bits	Purpose
Version	4	IP version = 4
IHL (header length)	4	Header length in 4-byte words; minimum 5 (20 bytes), maximum 15 (60 bytes)
Type of service / DSCP	8	Priority and quality of service
Total length	16	Header + data in bytes; maximum 65 535
Identification	16	Same value for all fragments of one datagram
Flags	3	DF (do not fragment), MF (more fragments follow)
Fragment offset	13	Position of the fragment's data, in units of 8 bytes
TTL (time to live)	8	Reduced by 1 at every router; at 0 the packet is dropped and ICMP "time exceeded" is sent
Protocol	8	Which protocol is inside: TCP = 6, UDP = 17, ICMP = 1
Header checksum	16	Error check on the header only; recomputed at each router
Source address	32	Sender's IP
Destination address	32	Receiver's IP
Options	variable	Rarely used

Fixed part = $4 + 4 + 8 + 16 + 16 + 3 + 13 + 8 + 8 + 16 + 32 + 32 = 160$ bits = **20 bytes**.

Fragmentation. Every link has an **MTU** (Maximum Transmission Unit; Ethernet = 1500 bytes). A datagram larger than the MTU is cut into **fragments**; each has its own IP header, and reassembly is done only at the **destination**. The data in each fragment (except the last) must be a **multiple of 8 bytes**.

Example. A datagram of total length 4000 bytes (20 header + 3980 data) crosses a link with MTU 1500. Data per fragment = $1500 - 20 = 1480$ bytes ($1480 \div 8 = 185$ ✓).

Fragment	Data bytes	Total length	MF flag	Offset (×8 bytes)
1	1480	1500	1	0
2	1480	1500	1	185
3	$3980 - 2960 = 1020$	1040	0	370

7.7 IPv6

● LEVEL 2 · EXAM

IPv6 uses **128-bit** addresses, giving $2^{128} \approx 3.4 \times 10^{38}$ addresses. They are written as **8 groups of 4 hexadecimal digits** (16 bits each) separated by colons.

Shortening rules: (1) drop leading zeros in each group; (2) replace **one** run of consecutive all-zero groups by **::** (only once in an address).

Example. 2001:0db8:0000:0000:0000:ff00:0042:8329 → 2001:db8:0:0:0:ff00:42:8329 → **2001:db8::ff00:42:8329**.

Address type	Prefix / example	Note
Global unicast	2000::/3	Public, routable on the Internet
Link-local	fe80::/10	Works only on the local link; every IPv6 interface has one
Unique local	fc00::/7	Like IPv4 private addresses
Multicast	ff00::/8	One-to-many (IPv6 has no broadcast)
Loopback	::1	Like 127.0.0.1
Unspecified	::	Like 0.0.0.0

Point	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal (192.168.1.1)	Hexadecimal with colons (2001:db8::1)

Point	IPv4	IPv6
Header	20–60 bytes, variable	Fixed 40 bytes, simpler
Header checksum	Yes	No (left to upper layers)
Fragmentation	Routers and sender may fragment	Only the sender fragments
Broadcast	Yes	No (uses multicast)
Configuration	Manual or DHCP	Auto-configuration (SLAAC) or DHCPv6
NAT needed	Often	No
Total addresses	About 4.3 billion	About 3.4×10^{38}

Moving from IPv4 to IPv6: **dual stack** (a device runs both), **tunnelling** (IPv6 packets inside IPv4) and **translation** (convert between them).

7.8 Routing basics

● LEVEL 2 · EXAM

A **router** chooses the next hop for every packet using its **routing table** (Chapter 3). The table can be filled by hand (**static routing**) or learned from other routers using **routing protocols** (**dynamic routing**). If several entries match, the router uses the **longest prefix match**: the most specific route wins.

Destination	Matching entries	Chosen
10.1.2.55	10.0.0.0/8, 10.1.0.0/16, 10.1.2.0/24 , 0.0.0.0/0	10.1.2.0/24 (longest, /24)
10.1.9.9	10.0.0.0/8, 10.1.0.0/16 , 0.0.0.0/0	10.1.0.0/16
8.8.8.8	only 0.0.0.0/0	Default route

Class	Idea	Protocols
Distance vector	Each router tells neighbours its whole table (distance to every network); uses the Bellman-Ford idea. Slow to react; can suffer count-to-infinity .	RIP (hop count, maximum 15, 16 = unreachable, update every 30 s, UDP 520)
Link state	Each router floods the state of its own links; every router builds a full map and runs Dijkstra's shortest-path algorithm. Faster and more accurate.	OSPF (cost based on bandwidth, uses areas)
Path vector	Routes between different organisations (autonomous systems), with policy control	BGP (TCP port 179); the routing protocol of the Internet

IGP vs EGP. An **autonomous system (AS)** is a group of networks under one administration. **Interior** protocols (RIP, OSPF) work **inside** an AS; the **exterior** protocol (BGP) works **between** ASes.

Distance-vector update example. Router B is 2 hops from A. A reports it is 5 hops from network D. So B can reach D via A in $2 + 5 = 7$. If B's current cost to D is 9, B **updates to 7 via A**.

QUICK TRICK

RIP = hop count, small networks, max 15 hops. **OSPF** = link state, Dijkstra, large networks inside an organisation.
BGP = between ISPs, the glue of the Internet.

7.9 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Forgetting – **2** in usable hosts, or giving 2^h instead.
- Calling 127.x.x.x a class A host address. 127 is **loopback**, not for the network.
- Using 172.0–172.255 as private. Only **172.16–172.31**.
- Writing a subnet mask such as 255.255.255.100: a valid mask has all 1s first then all 0s (octet values 128, 192, 224, 240, 248, 252, 254, 255).
- Using :: more than once in an IPv6 address.
- Starting VLSM with the smallest subnet. Always allocate the **largest first**.
- Treating a fragment offset as bytes. It is in units of **8 bytes**.

Quick-Solve Tricks — Summary

Situation	Trick
Class	1-126 A · 128-191 B · 192-223 C · 224-239 D · 240-255 E
Default masks	A /8 · B /16 · C /24
Hosts per subnet	$2^h - 2$; networks $2^b \geq$ needed
Block size	256 – mask octet; network = round host octet down to a multiple of block
Network / broadcast	Host bits all 0 / all 1
Private	10/8 · 172.16/12 · 192.168/16
Special	127.0.0.1 loopback · 169.254.x.x APIPA · 0.0.0.0 default
VLSM	Sort requirements largest first, allocate in order
Aggregation	Combine contiguous blocks; prefix shortens ($4 \times /24 \rightarrow /22$)
IPv4 header	20 bytes minimum, TTL, protocol TCP 6 / UDP 17 / ICMP 1
Fragment	Data multiple of 8; offset in 8-byte units; MF = 0 on last
IPv6	128 bits, hex, :: once, no broadcast, header 40 bytes
Routing	Longest prefix match; RIP (hop count 15), OSPF (Dijkstra), BGP (between ASes)

BPSC QUICK CHECK — Chapter 7

IPv4 → 32 bits, IPv6 → 128 bits	Class C default mask → 255.255.255.0
127.0.0.1 → loopback	Class D → multicast (224–239)
/26 → 62 hosts, block 64	/27 → 30 hosts, block 32
NAT → private to public	Longest prefix match picks the route
RIP → max 15 hops	OSPF → link state

Traps: hosts – 2 · 172.16–172.31 · broadcast and network addresses are not hosts · largest-first in VLSM · :: once only in IPv6.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** How many bits are in an IPv4 and an IPv6 address?
- **Q2.** Which class does the address 150.20.5.1 belong to?
- **Q3.** What is the default subnet mask of a Class C network?
- **Q4.** What is the loopback address?
- **Q5.** Write the three private IPv4 ranges.
- **Q6.** What does NAT do?

● LEVEL 2 · EXAM

- **Q7.** Convert 192.168.1.10 to binary.
- **Q8.** Convert 10101100.00010000.00001010.00000001 to dotted decimal.
- **Q9.** For 192.168.5.77/26 find the network address, broadcast address, usable host range and number of hosts.
- **Q10.** How many usable hosts are there in a /27, a /29 and a /30 network?
- **Q11.** The Class B network 172.16.0.0 is to be split into at least 8 subnets. Find the new mask and the hosts per subnet.
- **Q12.** Write 2001:0db8:0000:0000:0000:0000:0001 in its shortest form.

● LEVEL 3 · BPSC / PGT

- **Q13.** Divide 192.168.20.0/24 into 8 equal subnets. Give the mask, hosts per subnet, and the network and broadcast address of the first three subnets.
- **Q14.** Using VLSM on 192.168.50.0/24, allocate subnets for LAN1 = 60 hosts, LAN2 = 28 hosts, LAN3 = 12 hosts and two WAN links of 2 hosts each.
- **Q15.** A datagram of total length 3000 bytes (20 header + 2980 data) must cross a link with MTU 1000. Find the number of fragments, each fragment's data length, total length, MF flag and offset.
- **Q16.** Aggregate 200.1.4.0/24, 200.1.5.0/24, 200.1.6.0/24 and 200.1.7.0/24 into one prefix.
- **Q17.** A router has the routes 10.0.0.0/8 → R1, 10.1.0.0/16 → R2, 10.1.2.0/24 → R3 and 0.0.0.0/0 → R4. Where does it send packets for 10.1.2.55, 10.1.9.9, 10.9.9.9 and 8.8.8.8?
- **Q18.** Find the network address, broadcast address and number of usable hosts for 10.1.37.200/21.
- **Q19.** Router B is 2 hops from A. A advertises 5 hops to network D. B's current cost to D is 9. What does B do? What is the largest valid hop count in RIP?
- **Q20.** In the subnet 192.168.10.64/26, which of these are valid host addresses: .64, .65, .126, .127? How many addresses does a /22 contain in total?

Answers

Q	Solution
1	IPv4: 32 bits. IPv6: 128 bits.
2	First octet 150 is in 128–191, so Class B .
3	255.255.255.0 (/24).
4	127.0.0.1 (the whole 127.0.0.0/8 block is loopback).
5	10.0.0.0/8 (10.0.0.0–10.255.255.255), 172.16.0.0/12 (172.16.0.0–172.31.255.255), 192.168.0.0/16 (192.168.0.0–192.168.255.255).
6	A NAT router translates private addresses to a public address (and back), so many hosts share one public IP.
7	11000000.10101000.00000001.00001010 .
8	10101100 = 172, 00010000 = 16, 00001010 = 10, 00000001 = 1, so 172.16.10.1 .
9	/26 → block size 64. 77 → 64. Network 192.168.5.64 , broadcast 192.168.5.127 , hosts 192.168.5.65 - 192.168.5.126 , $2^6 - 2 = 62$ hosts.
10	/27: $2^5 - 2 = 30$. /29: $2^3 - 2 = 6$. /30: $2^2 - 2 = 2$.
11	$2^3 = 8$, borrow 3 bits: /16 + 3 = /19 , mask 255.255.224.0 . Host bits = 13, hosts = $2^{13} - 2 = 8190$.
12	2001:db8::1 .
13	Borrow 3 bits: /27 , mask 255.255.255.224 , block 32, hosts $2^5 - 2 = 30$. Subnet 1: 192.168.20.0, broadcast .31. Subnet 2: .32, broadcast .63. Subnet 3: .64, broadcast .95. (...last: .224, broadcast .255.)
14	Sort: 60 → /26 (62), 28 → /27 (30), 12 → /28 (14), WAN → /30. LAN1 192.168.50.0/26 (0–63); LAN2 192.168.50.64/27 (64–95); LAN3 192.168.50.96/28 (96–111); WAN1 192.168.50.112/30 (112–115); WAN2 192.168.50.116/30 (116–119). 192.168.50.120 onward is free.
15	Max data = 1000 – 20 = 980, rounded down to a multiple of 8 = 976 . Fragments: 976, 976, 976 and 2980 – 2928 = 52, so 4 fragments . Total lengths 996, 996, 996, 72. MF = 1, 1, 1, 0. Offsets = 0, 122, 244, 366.
16	Third octets 4–7 = block of 4 starting at a multiple of 4, so 200.1.4.0/22 .
17	10.1.2.55 → R3 (/24). 10.1.9.9 → R2 (/16). 10.9.9.9 → R1 (/8). 8.8.8.8 → R4 (default).
18	Mask /21 = 255.255.248.0, block 8 in the third octet; 37 → 32. Network 10.1.32.0 , broadcast 10.1.39.255 , hosts $2^{11} - 2 = 2046$.
19	Cost via A = 2 + 5 = 7 < 9, so B updates to 7 via A . Largest valid RIP hop count = 15 (16 means unreachable).
20	Network .64 and broadcast .127 are not hosts; valid hosts are .65 and .126 . A /22 has $2^{10} = 1024$ addresses (1022 usable).

REMEMBER

Next: Chapter 8 — Web Services (WWW, web page and website, URL, web browser and web server, HTML basics, XML, domain names, web hosting, search engines, e-commerce basics).

Roadmap: Ch 9 Network Security (threats, malware, firewall, cryptography, digital signature, cyber laws) · Ch 10 Mobile Telecom (1G to 5G), Cloud and IoT.